

Joseph Rabara Bailey

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Research-focused Stanford BS/MS student working at the intersection of fundamental physics, machine learning, and statistical inference; NCAA Division I hurdler and musician with interests in AI for scientific discovery, neutrino physics, high-energy physics, and secure systems.

EDUCATION

Stanford University, Stanford, CA (*Expected June 2027*)

Coterminal B.S. in Physics and M.S. in Computer Science. Undergraduate cumulative GPA: 3.980; graduate cumulative GPA: 3.900 (June 20, 2026 unofficial transcript).

- Selected coursework: Natural Language Processing with Deep Learning, Machine Learning with Graphs, Deep Learning for Computer Vision, Operating Systems, Cryptography, Theory of Computation, Data Structures and Algorithms, Information Theory, Quantum Mechanics, Electrodynamics, Advanced Mechanics, General Relativity, Statistical Mechanics, Real Analysis, Probability Theory, Partial Differential Equations, and Scientific Communication in Physics.
- Planned 2026-27 coursework: Quantum Field Theory and Reinforcement Learning.
- AP credit: Chemistry, Calculus BC, and Physics 1/2.

Chino Hills High School, Chino Hills, CA (*Class of 2023*)

GPA: 4.47. National Merit Finalist and ACS U.S. National Chemistry Olympiad National Finalist.

CURRENT ACADEMIC DIRECTION

- Primary academic focus: physics-informed machine learning, statistical inference, and AI methods for extracting physical measurements from noisy, incomplete detector data.
- Current honors-thesis research direction: benchmark OmniFold, an unbinned machine-learning unfolding framework, against iterative Bayesian unfolding on MINERvA neutrino scattering data; evaluate closure tests, generator stress tests, multidimensional correlations, covariance stability, and bias-variance tradeoffs.
- Longer-term goal reflected in application materials: pursue a Ph.D. focused on AI for fundamental science.

RESEARCH EXPERIENCE

Undergraduate Researcher, Machine Learning-Based Unbinned Unfolding for Neutrino-Nucleus Cross Section Measurements, Stanford Physics, Prof. Benjamin Nachman (*Stanford, CA | Winter 2026-Present*)

- Implementing and evaluating OmniFold on MINERvA open data to test whether high-dimensional, unbinned unfolding can reduce structural model bias in neutrino cross-section extraction.
- Comparing machine-learning unfolding with iterative Bayesian unfolding using closure tests, generator stress tests, bootstrap resampling, covariance comparisons, and principal-component structure analysis.
- Developing this work toward an honors thesis chapter and manuscript outline.

Researcher, SLAC National Accelerator Laboratory (*Menlo Park, CA | June 2024-Jan. 2025*)

- Collaborated on a four-person team proposing a particle collider based on gamma-gamma collision.
- Used Delphes, ROOT, and Unix-based high-performance computing workflows to simulate and analyze terabytes of particle-collider data.
- Applied machine-learning methods to study the significance of pileup/background particles on detection of di-Higgs events.

Research Intern, Caltech LIGO Lab (*Pasadena, CA | June 2023-July 2023*)

- Assisted with simulation, testing, and preparation of suspension elements for Laser Interferometer Gravitational-wave Observatory runs.
- Conducted experiments on laser-beam properties in an interferometer and used Python for data analysis.

PROFESSIONAL AND ENGINEERING EXPERIENCE

Product Management Intern, Digital Assets Team, WisdomTree Digital Movement, Inc. (*New York, NY | June 2026-Present*)

- Supporting product management, market research, internal documentation, and product-development discussions for digital-asset financial products.
- Role bridges technical interests in blockchain/digital assets with financial markets and client-focused product strategy.

Engineering Intern, Polytec (*Pasadena, CA | June 2023-Sept. 2023*)

- Designed statistical tools for customers analyzing vibration data with Python, MATLAB, and Octave.
- Modeled and 3D-printed laboratory tools to streamline operations.

SELECTED TECHNICAL AND RESEARCH PROJECTS

SPLoRA: Parameter-Efficient Fine-Tuning of GPT-2 with LoRA

Implemented LoRA for GPT-2 for Quora paraphrase detection and Shakespearean sonnet generation; ran ablations over adapter placement and rank/scaling hyperparameters; achieved 0.888 development accuracy while training roughly 1% of model parameters and 41.94 chrF on sonnet generation.

Person Re-Identification with Vision Foundation Models

Current CS 231N project using vision foundation models, LoRA adapters, and temporal attention over video tracklets for person re-identification.

Soccer Action Prediction with Graph Neural Networks

Modeled soccer possessions as heterogeneous graphs using Wyscout data; trained GraphSAGE and Relational Graph Transformer models with RelBench and PyG to predict best next action.

Gaussian Process Bump Hunting

Developed a codebase for Jefferson Lab to search for the hypothetical X17 particle using Gaussian-process regression to detect bumps/excesses in particle-collider data; presented at the 2025 APS Far West Section Annual Meeting at UC Santa Cruz.

On-Chain Ethereum Wallet

Built a programmable wallet in Solidity and Foundry supporting ETH and ERC-20 transfers, approvals, deposits, withdrawals, and a 2-of-3 multisig authorization scheme.

Pintos OS Enhancements

Implemented thread scheduling, system calls, user programs, virtual memory, and file-system support for a primitive x86 operating system using C, kernel programming, and Git.

End-to-End Encrypted Chat Client

Implemented the Double Ratchet Algorithm, based on protocols used in Signal and WhatsApp, for secure messaging.

Keychain: Secure Password Manager

Built a credential manager using AES-GCM authenticated encryption and SHA-256 hashing for secure credential storage.

2nd Chance: Generative Conversational AI

Engineered a system that synthesized conversations by emulating text and voice patterns from provided logs using ChatGPT, Whisper, and ElevenLabs APIs.

Sports Win Prediction with Machine Learning

Researched machine-learning algorithms using only pre-match data to predict professional League of Legends match outcomes with reported 90% accuracy.

PRESENTATIONS, GRANTS, AND RESEARCH OUTPUTS

- Presenter, Gaussian Process Bump Hunting, APS Far West Section Annual Meeting, University of California, Santa Cruz, 2025.
- Honors thesis and manuscript in preparation: systematic evaluation of OmniFold in the few-GeV neutrino regime using MINERvA open data.

AWARDS AND HONORS

- Stanford VPUE Major Grant, 2026 (awarded after external review).
- Stanford Track and Field Heart and Hustle Award, 2025.
- National Merit Finalist.
- ACS U.S. National Chemistry Olympiad National Finalist.
- Music Teachers' Association of California Certificate of Merit, Piano Level 10; State Honors and Branch Honors.
- Chino Hills High School: Best All-Around Male Student and Homecoming King.
- Black Belt in Karate.

LEADERSHIP, ATHLETICS, SERVICE, AND MUSIC

Student-Athlete, Stanford Track and Field, NCAA Division I (*Stanford, CA | Jan. 2024-Present*)

- Hurdler and sprinter, including 400-meter hurdles, training more than 20 hours per week while carrying a demanding Physics and Computer Science course load.

Undergraduate Representative, Physics Department Undergraduate Studies Committee (*Stanford, CA | Sept. 2025-June 2026*)

- Served as the undergraduate voice on the department committee shaping the physics curriculum, course offerings, and advising; met monthly to review the undergraduate program and propose improvements.

Dorm Co-President, Stanford Freshman Dorm (*Stanford, CA | Sept. 2023-June 2024*)

- Helped lead a 90-person dorm community through weekly meetings, event planning, peer engagement, and community standards.

Musician, Stanford Jazz Orchestra and Stanford Jazz Combos (*Stanford, CA | Sept. 2023-Jan. 2024*)

- Performed alto saxophone, flute, and clarinet in Stanford's top jazz ensemble and jazz combos.

Section Leader, Jazz Captain, and Saxophone Soloist, Chino Hills High School Marching Band, Drumline, and Jazz Band (*Chino Hills, CA | Aug. 2019-Apr. 2023*)

- Led alto saxophone sectionals, served as jazz captain, and performed as marching-band saxophone soloist for three championship seasons.
- Contributed to three state-championship marching-band seasons; first alto saxophone for four years; self-taught flute, clarinet, and soprano saxophone.

- Treasurer and Tutor**, High School Health Science Academy (*Chino Hills, CA | Aug. 2019-May 2023*)
- Managed finances for fundraisers and clothing sales, supervised blood-drive logistics, and tutored students during lunch.
- Central Supply Volunteer**, Casa Colina Hospital (*Pomona, CA | Jan. 2022-Sept. 2022*)
- Supported hospital supply operations as a volunteer.

TECHNICAL SKILLS

Area	Tools and Methods
Programming	Python, C/C++, Java, JavaScript, Solidity, MATLAB, Octave
Machine Learning	PyTorch; CNNs, RNNs, LLMs, and vision transformers; LoRA; graph neural networks; OmniFold; Gaussian processes; boosted trees
Physics and Data Analysis	ROOT, Delphes, detector simulation, unfolding, data augmentation, bias/dark/flat-field correction, differential photometry, periodograms, error propagation, covariance analysis, high-performance Unix computing
Systems and Security	Operating systems, kernel programming, cryptography, AES-GCM, SHA-256, Double Ratchet, Ethereum smart contracts, Foundry
Languages and Communication	First-year Modern Chinese coursework; scientific writing and communication coursework

RESEARCH INTERESTS

- AI for fundamental science; scientific machine learning; particle and neutrino physics; uncertainty quantification; interpretable statistical inference; secure systems; digital assets and financial technology.